



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Lecture 5: CA6121 AI in Project Management

Project Procurement, Quality, Risks, Streamlined Work Processes, Productivity

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OBJECTIVES

To learn and understand:

- Overview of Project Procurement
- Procurement process cycle
- Procurement risks and AI tools
- Project Quality Overview
- Project Risks Overview
- Case Studies: AI tools



Project Procurement

Project Procurement



Definition

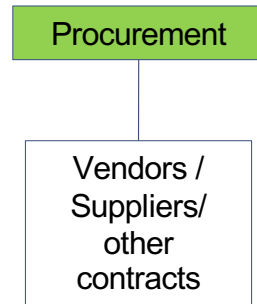
What is Procurement?

Project Procurement Management includes the processes necessary to **purchase or acquire products, services**, or results needed from outside the project team.

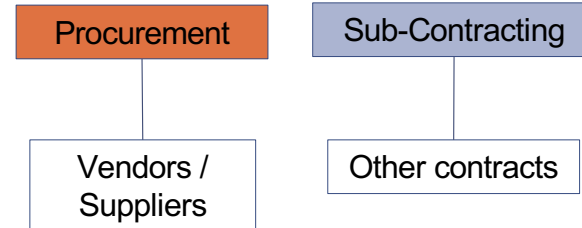


Procurement

Some companies use terminology
“**Procurement**” only for purchasing of
Equipment and other items

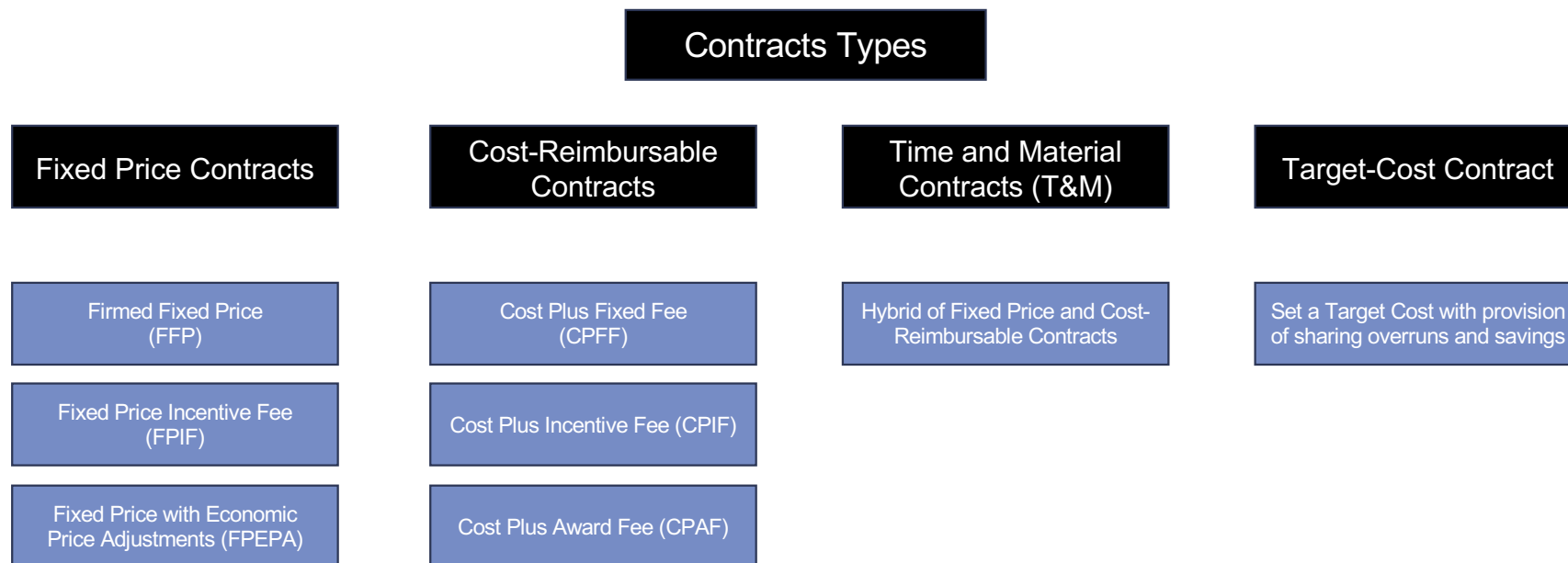


Some companies use terminology
“**Procurement**” to combine purchasing of
Equipment and other items PLUS sub-
contracts





Plan Procurement Management





Procurement Process Flow

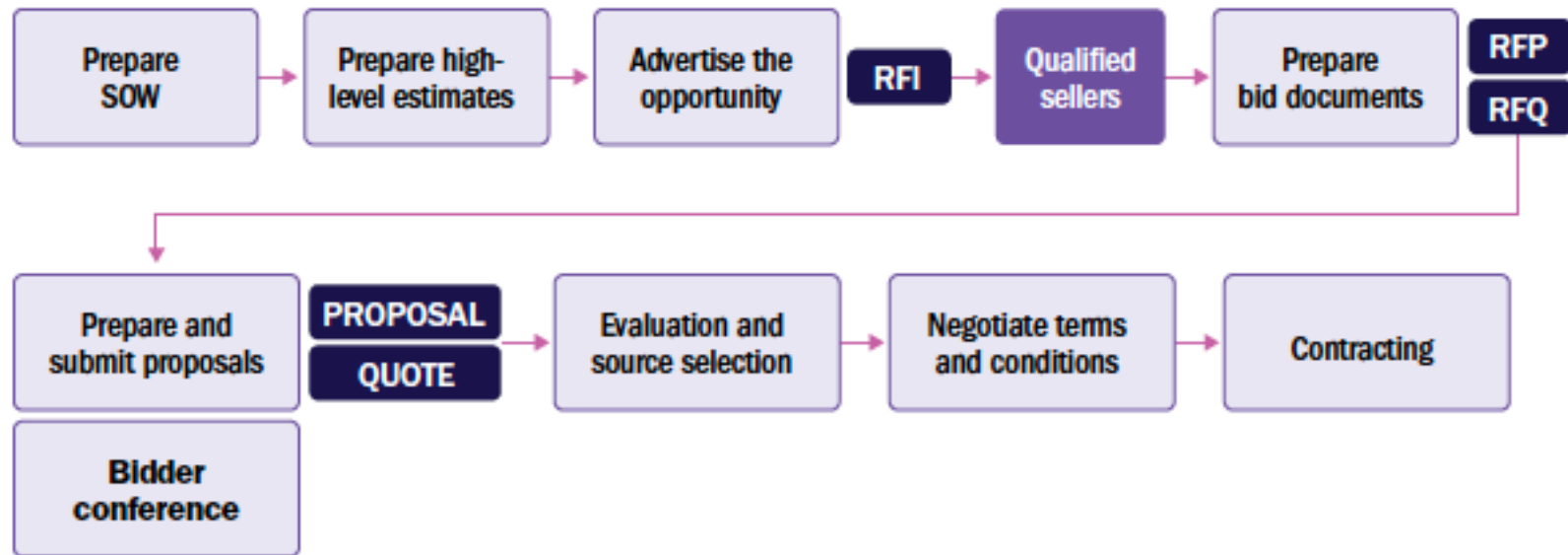


Figure X4-1. Example Procurement Process Flow

SOW: Schedule of Works
RFI: Request for Information
RFP: Request for Proposal
RFQ: Request for Quote



Procurement Management Process

1. Plan Procurement Management

2. Conduct Procurements

3. Control Procurements



Plan Procurement Management

Inputs

- 1. Project Charter**
- 2. Business Documents**
 - Business Case
 - Benefits management plan
- 3. Project Management Plan**
 - Scope Management Plan
 - Quality Management Plan
 - Resource Management Plan
 - Scope Baseline
- 4. Project Documents**
 - Milestone List
 - Project Team Assignments
 - Requirements Traceability Matrix
 - Resource Managements
 - Risk Register
 - Stakeholder Register



Plan Procurement Management

Inputs

5. Enterprise Environmental Factors

- Marketplace conditions
- Availability of product or services in the marketplace
- Seller past performance and reputation
- Terms and Conditions applicable
- Local and regulatory requirements
- Legal requirements
- Contract management system
- Established multi-tier supplier system
- Financial Accounting and Contract Payments System

6. Organizational process assets

- Approved seller lists
- Formal Policies, Procedures and Guidelines
- **Contract Types**



Plan Procurement Management

Tools & Techniques

- a. Expertise and Past Experience**
- b. Data Gathering**
 - Market Research
- c. Data Analysis**
 - Make-or-buy analysis
- d. Source Selection Analysis**
 - Least Cost
 - Qualifications Only
 - Quality Based / Highest Technical Proposal Score
 - Quality and Cost Based
 - Sole Source
 - Fixed Budget
- e. Meetings**



Plan Procurement Management

Outputs

a. Procurement Management Plan

- Project procurement activities coordination
- Key Procurement Activities Schedule
- Role & Responsibilities
- Legal Requirements
- Estimates
- Risk Management
- Pre-qualified sellers

b. Procurement Strategy

- Delivery Methods
- Contract payment types (Fixed, Cost plus, incentives etc)
- Procurement Phases



Plan Procurement Management

Outputs

c. Bid Documents

- Request for Information
- Request for quotation
- Request for proposal

d. Procurement Statement of Work

e. Source Selection criteria

- Capability and Capacity
- Product cost and life cycle cost
- Delivery dates
- Technical expertise
- Experience
- Comply to specifications and requirements
- Resources and their suitability and competence
- Financial Stability



Plan Procurement Management

Outputs

- f. Make-or-buy decisions**
- g. Independent Cost Estimates**
- h. Change Requests**
- i. Project Documents Updates**
 - Lessons/ Risk / Stakeholder Register
 - Requirement's documentation
- j. Organizational Process Assets Updates**



Conduct Procurement

Outputs

a. Select Sellers

b. Agreements

- Scope
- Schedule
- Performance progress reporting
- Pricing and Payment terms
- Inspection, Quality and Acceptance criteria
- Warranty
- Incentives and penalties
- Insurance and performance bonds
- General T&C
- Termination clause and dispute resolution clause

c. Change Requests



Conduct Procurement

Outputs

d. Project Management Plan Updates

- Requirements updates
- Quality Management Plan
- Communications management plan
- Risk Management Plan
- Procurement Management Plan
- Scope baseline
- Schedule baseline
- Cost baseline



Conduct Procurement

Outputs

e. Project Documents Updates

- Lessons Register
- Requirements documentation
- Resource calendars
- Risk Register
- Stakeholder Register

f. Organizational Process Assets Updates



Control Procurement

Outputs

- a. Closed Procurements**
- b. Work Performance Information**
- c. Procurement Documentation Updates**
- d. Change Requests**
- e. Project Management Plan Updates**
 - Risk Management Plan
 - Procurement Management Plan
 - Schedule Baseline
 - Cost Baseline



Control Procurement

Outputs

f. Project Documents Updates

- Lessons Register
- Resource Requirements
- Risk Register
- Stakeholder Register

g. Organizational Process Assets Updates



Project Success Factors

Project Procurement Management



Assessing Project Success

2 Dimensions

Effectiveness of Project
in realizing the intended
value.

Efficiency of Project
Management Processes



Project Success Factors

- Timely and good quality of:
 - Overall scope and concept deliverables
- Realist achievable schedule
- Timely:
 - Deliverables from Engineering: Material/Contract requisition, standards, specs.
 - Issuance of request for quote.
 - Bid reviews thru final technical and commercials.
 - Placement of Purchase orders for equipment, bulk items, material.
 - Placement of Sub-contracts for installation and other services.
 - Fabrication.
 - Inspection.
 - Factory Acceptance Test for equipment.
 - Shipping and delivery on time.
 - Site acceptance, inspection.



Project Success Factors

- Good planning and monitoring
- Good Material Control and Management
- Installation quality
 - Development / Fabrication / Construction monitoring
 - Installation
- Minimize rework



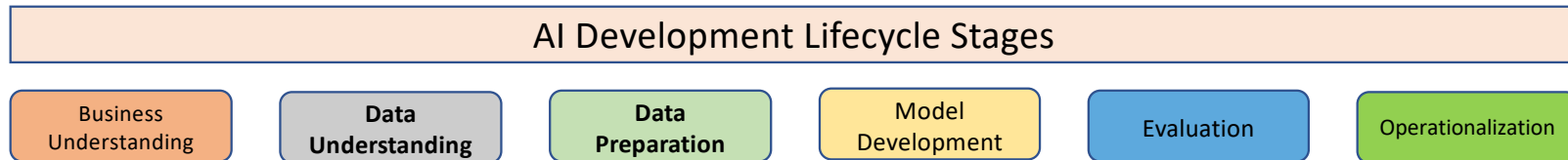
Procurement Process Cycle

Project Procurement Management



AI Development Projects

Recap

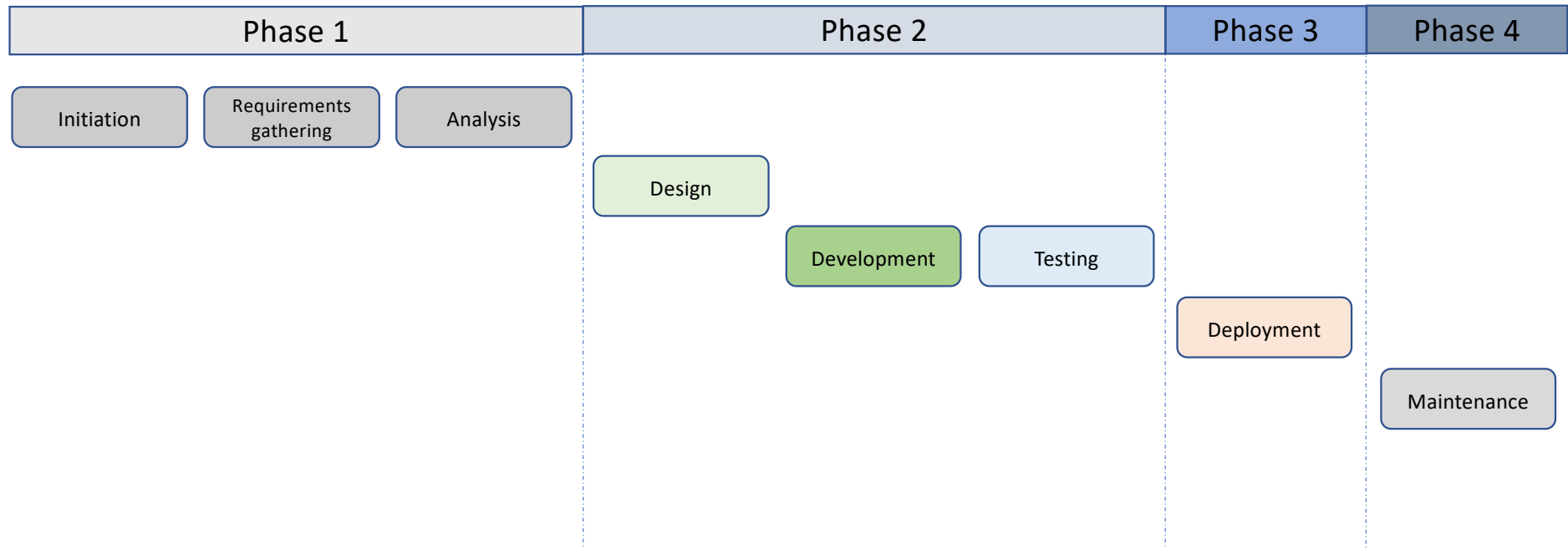


Project Manager could be from either of the disciplines:

- Computer Science
- Data Science
- Information Technology
- Other discipline with knowledge and interest in data

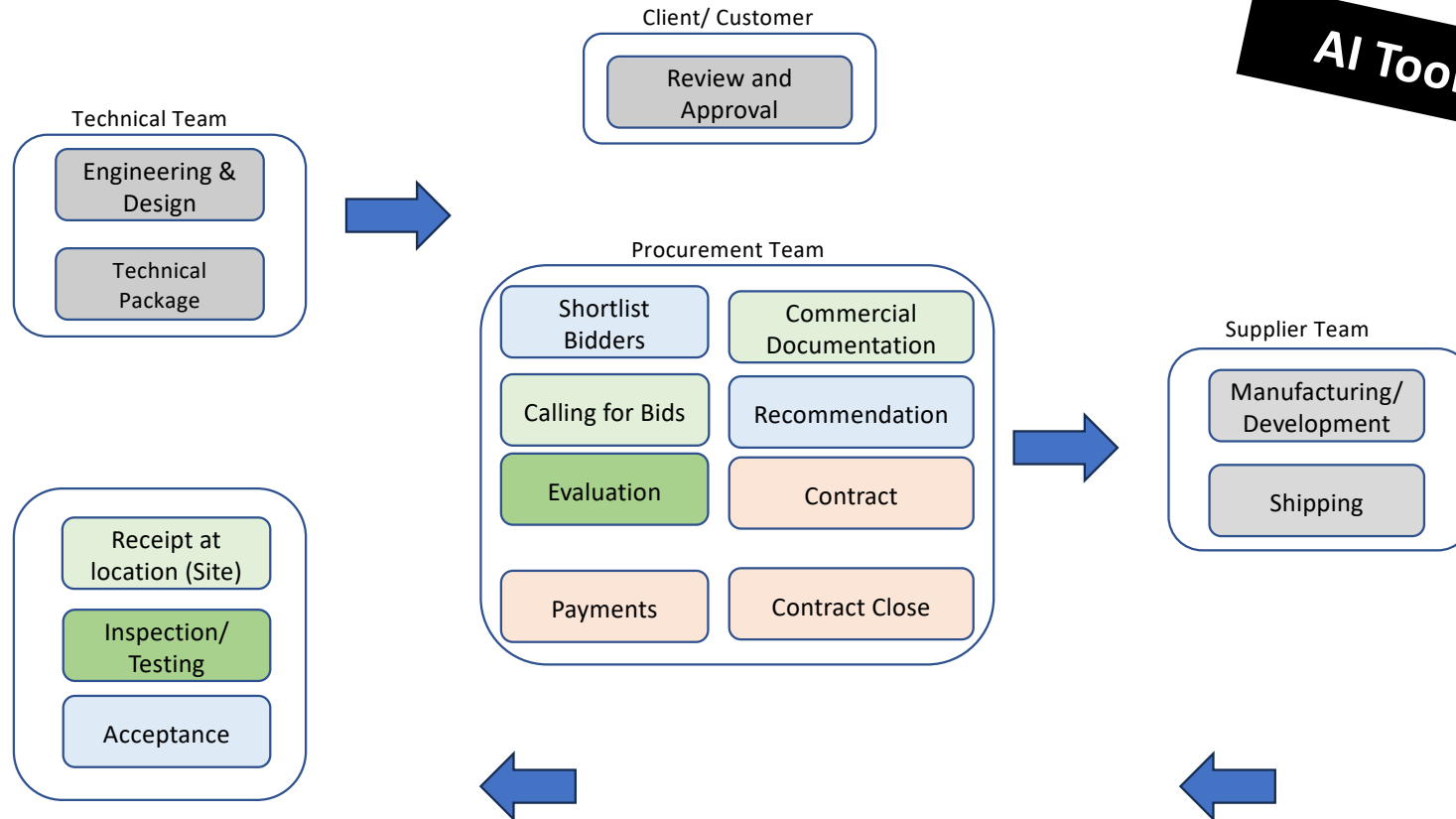


Technology/ Software Development Projects





Procurement Process Cycle



AI Tool can help



Procurement: Pre-Order and Post Order Activities

AI Tool can help

Procurement Prime Responsibility:

Purchase Equipment, items, place orders and ensure its timely delivery to the project

Procurement Activities

Pre-Order/Award Activities

- Vendor Selection
- Preparing Bid / Tender Documents
- Calling Bids/ Tenders
- Evaluating
- Award

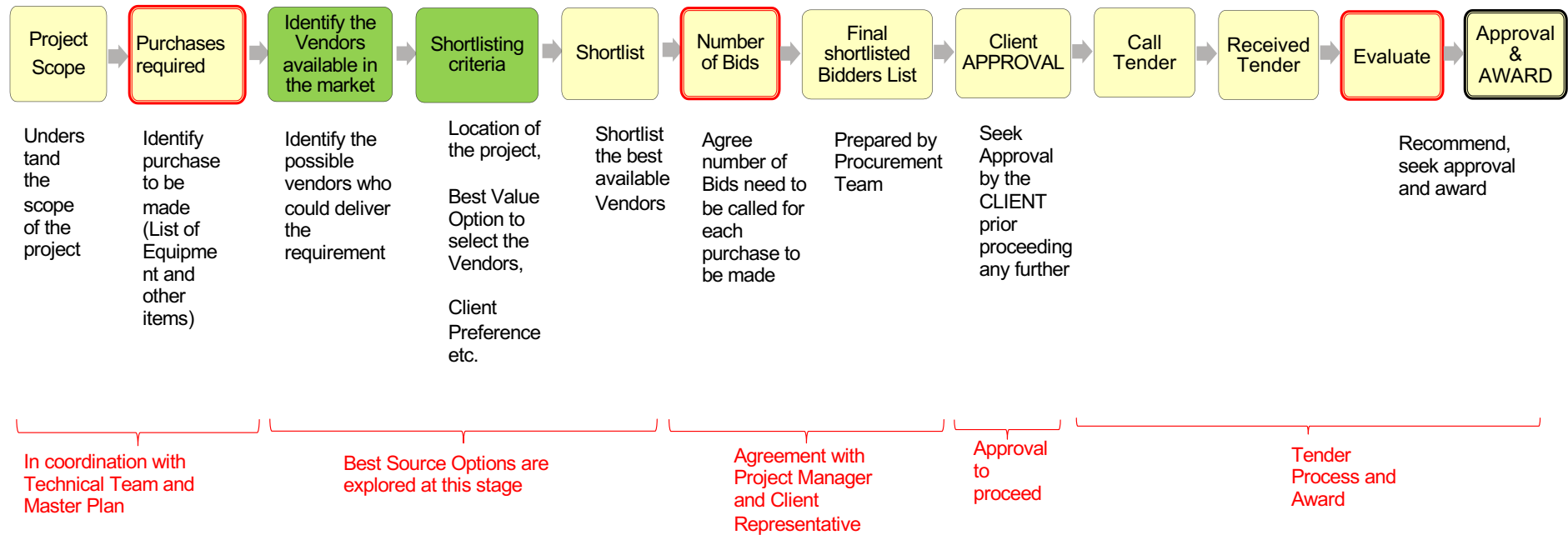
Post-Order/Award Activities

- Order tracking and coordination
- Expediting and Inspection
- FAT (Factory Acceptance Test) witness
- Shipping and Delivery to the site
- Material Control and Management
- Over, Short and Damaged (OS&D)
- Contract close out



Procurement Process Cycle

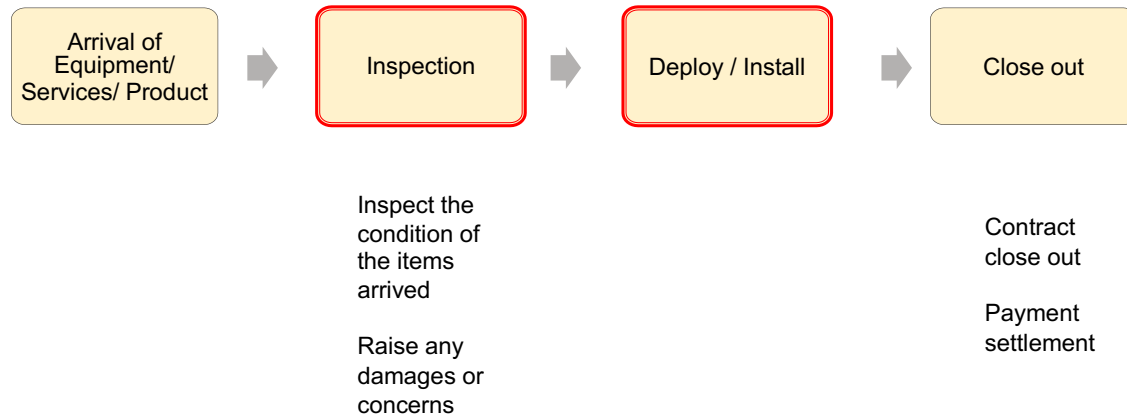
**Pre-Order
Activities**





Procurement Process Cycle

**Post-Order
Activities**





Procurement Risks

Project Procurement Management



Procurement Risks: Vendors / Suppliers / Sub-Contractors

AI Tool can help

Timely Approvals:

- Can we proceed without client's approvals?
- Any client suggested vendor/ supplier/ sub-contractor/ specialist must be recorded and duly signed by the client.
- Such list must be mentioned in PMP (Project Management Plan).
- Past performance and financial checks still must be performed on client's suggested vendor/ supplier/ sub-contractor/ specialist.
- Single source items and vendor/ supplier/ sub-contractor/ specialist must be recorded and duly signed.



Procurement Risks

Risks need to be managed:

- Financial failure of the Vendor / Supplier/ Sub-contractor
- Under performance
- Delays (Development, Fabrication, Delivery, Installation etc.)

AI Tool can help



Project Quality

Project Quality Management



Definition

What is Quality???



Definition

Client Asked for GOLD Plated Item

1. If you provide him/her a **Platinum** Plated...
Is it **QUALITY**???
2. If you provide him/her a **Silver** Plated...
Is it **QUALITY**???
3. If you provide him/her a **Gold** Plated...
Is it **QUALITY**???





Definition

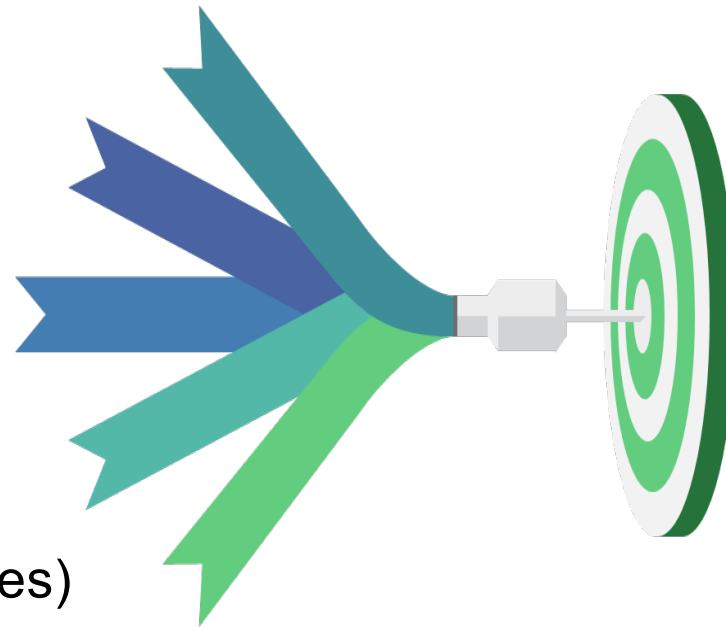
Meeting customer's expectations and perceptions





What is Quality?

- Fit for the purpose
- Perception of the end-user
- Customer focus
- Concepts:
 - Plan
 - Manage
 - Control
- Quality Assurance (Project Processes)
- Quality Control (Project Deliverables)





Quality Management

Quality Management processes

include all the activities that

determine quality policies, objectives and responsibilities

so that the project will

satisfy the quality requirements and produce a product

that meets quality standards.



Project Quality Management

International Organization of Standardization (ISO)

Most Common Standards

ISO 9000	Quality Management System Standard
ISO 14000	Environment Management
ISO 27000	Information Security Management
ISO 31000	Risk Management
ISO 45000	Occupational Health & Safety

Fun Fact: Which ISO Standard defines currency?

ISO 4217 Currency Codes



Project Quality Management

Modern PQM (Project Quality Management) approaches:

- a. Customer Satisfaction
- b. Prevention over Inspection
- c. Continuous Improvement (PDCA, TQM, Six Sigma, Lean Sigma)
- d. Management Responsibility
- e. Cost of Quality (COQ)



Consequence of Bad Quality

Defective goods/ products/ service delivery spoils image of the manufacturer/ developer/ service provider and leads to:

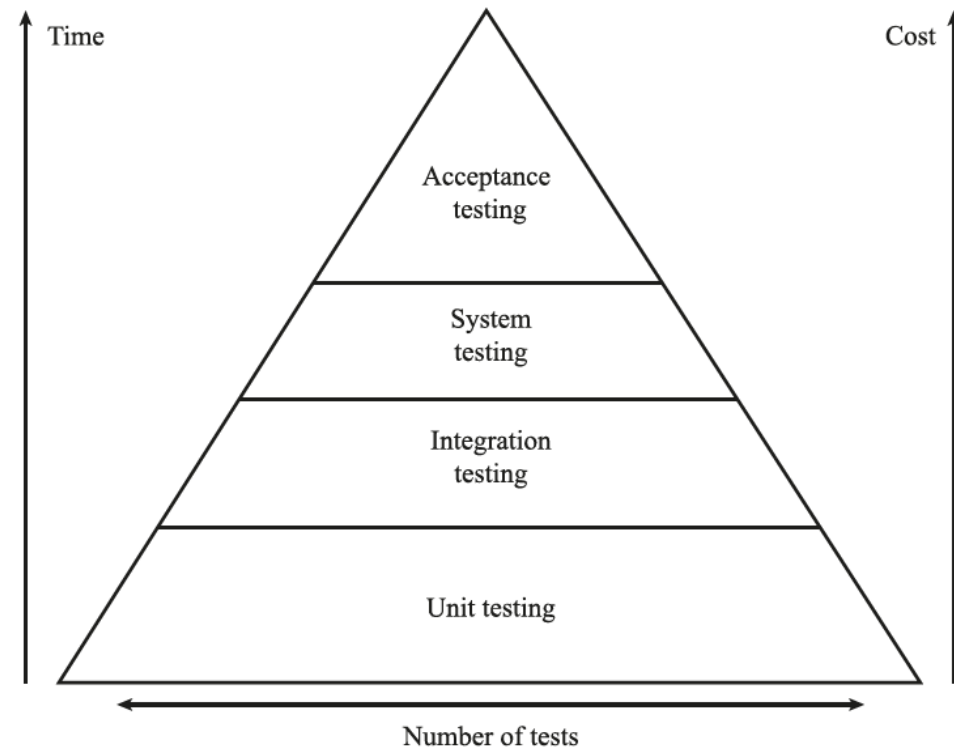
- Re-Work
- Resources Losses
- Efficiency and Productivity Losses
- Financial losses
- Reputation losses



Functional Requirements Testing Pyramid

Test types are ranked by:

- Importance
- Speed
- Cost
- Success rate





Type of Software Development Testing

- Unit testing, or modular testing
- Integration testing.
 - Big Bang
 - Incremental Approach
 - Bottom-up Integration
 - To-down Integration
 - Hybrid Integration
- System Testing



Type of Software Development Testing

- Acceptance Testing
 - Performance Testing
 - Load Testing
 - Fault Tolerance Testing
 - Compatibility Testing
 - Security Testing
 - Disaster Recovery Testing



Unit testing, or modular testing

- A check of the **correctness of individual modules** of the **program source code**, individual **functions** of the application or **service**.
- This establishes whether the **code is operational**, whether it **gives correct results with different inputs**, and whether the **logic matches the expected** one.
- An example:
 - Obtain the sum of two numbers
 - The input values required for testing are entered into the function
 - The unit test is passed if the expected result of the function is equal to the one actually obtained.
 - The metric is calculated by comparing the number of lines of code that passed the test to the total number of lines of code in the repository.
- The best indicator is considered to be 80%.



Integration Testing

- Testing the interaction of the components of the future application.
- The integration of modules is checked.
- These tests are more complex than unit testing, since the ability to perform sequential actions in various components is checked.
- Example:
The user logs in — is redirected to the main page — places an ad.
That is, several modules are tested at once.



System Testing

- All components of the program are tested as a single application.
- The specialist must ensure that the product correctly handles various scenarios and situations based on requirements.
- This option **may include testing non-functional requirements**.
- System testing requires considerable time, because as the product evolves, the number of possible use cases grows.
- With each product update, it is necessary to check whether the previously tested functions work. This process is called regression testing.



Acceptance Testing

- The product is tested by key specialists and customers.
-
- Check in real time
- Ensure expectations and requirements have been met.
-
- Acceptance is the final stage of testing.
- Minimize any bugs or defects fixing at this stage.

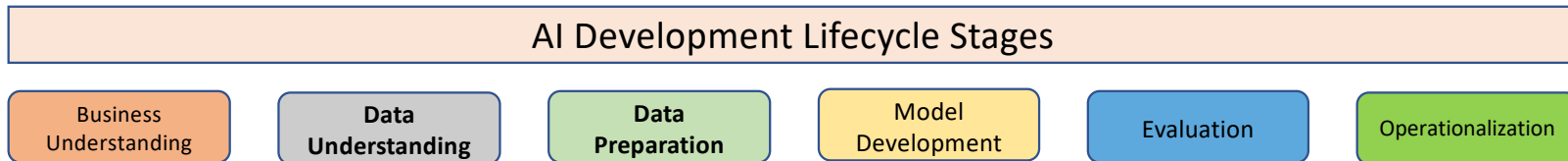
Non-functional testing **checks the application** for:

- performance
- reliability
- compatibility
- security
- utility
- scalability



AI Development Projects

Recap



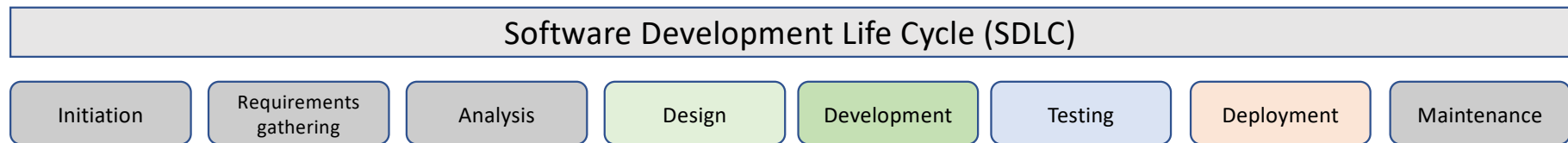
Project Manager could be from either of the disciplines:

- Computer Science
- Data Science
- Information Technology
- Other discipline with knowledge and interest in data

Quality assurance & Control is important



Quality Management Process in SDLC



Quality assurance is part of every step



Project Risk Assessment

Project Risk Management



Definition

What is Risk?

Project Risk is an **uncertain event or condition** that, if occurs,
has a positive or negative **effect on** one or more project
objectives such as
scope, schedule, cost and quality.

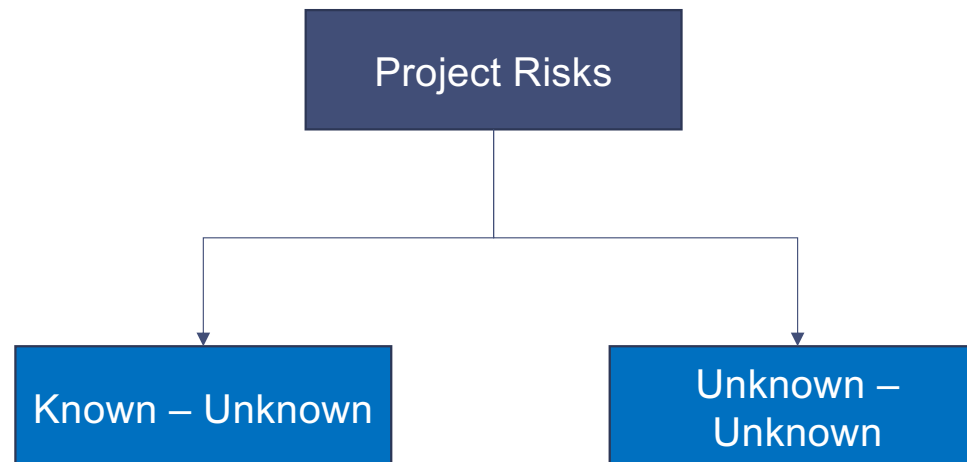


Risks

- Risks will continue to emerge through out project lifecycle.
- Risks must be monitored and managed.
- Important to manage the risks to deliver the project within the expectations of contractual agreement.
- Level of acceptable level of risk exposure to be agreed and known upfront.
- Contingencies are built in the cost and schedule estimates for risks.



Risks





Risk Classification

Unknown–Known (Hidden fact) Knowledge exists in the community but not with the entity working on the endeavor.	Unknown–Unknown (Emergent risk) Knowledge does not exist within the sphere of influence.
Known–Known (Facts and requirements) Managed as a part of scope. Not a risk.	Known–Unknown (Classic risk) There is knowledge to identify probability and impact.

Figure 2-46. Risk Classification



Risk Classification

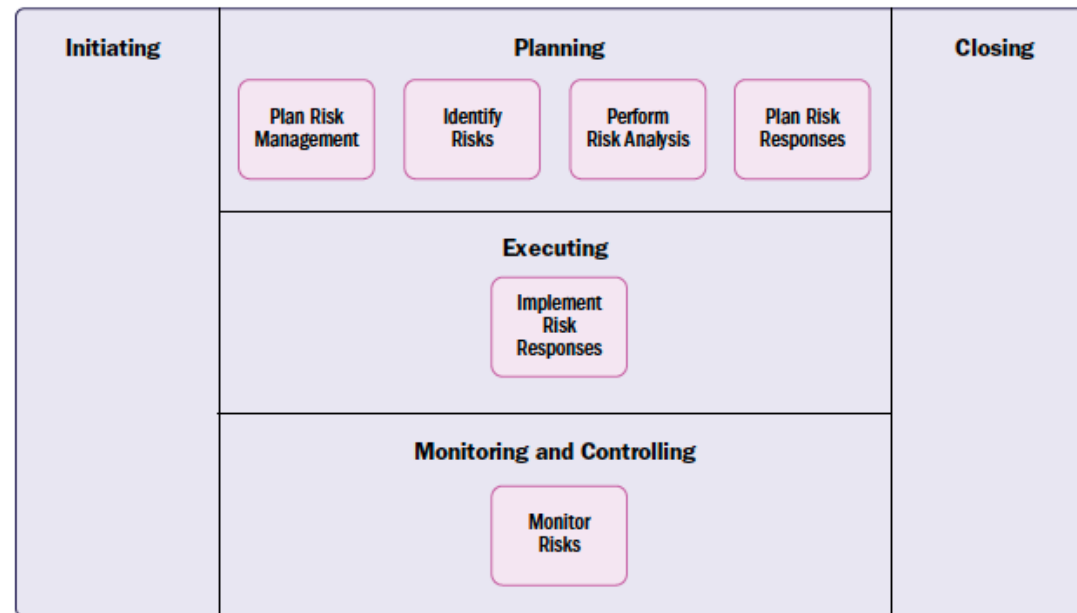
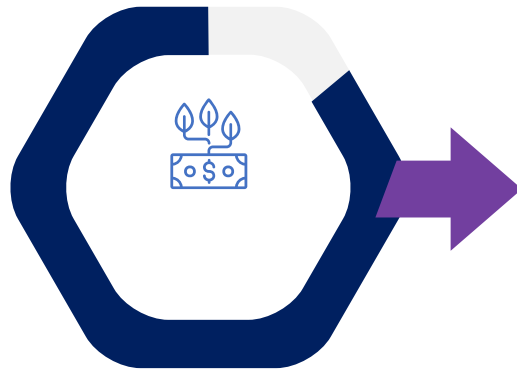


Figure 2-47. Risk Performance Domain Processes Overview



Level of Risks

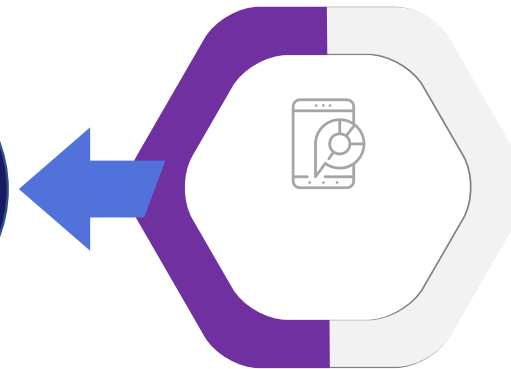
INDIVIDUAL PROJECT RISK



- Impact on one or more project objectives.

Risks

OVERALL PROJECT RISK



- Exposure to the project variations.
- Impact on overall project.



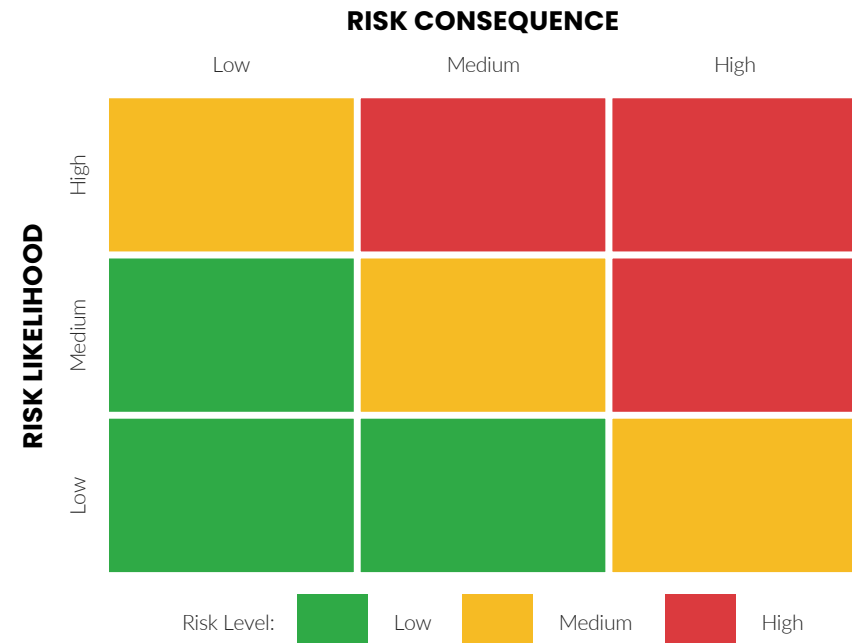
Risk Matrix

Project Risk Management



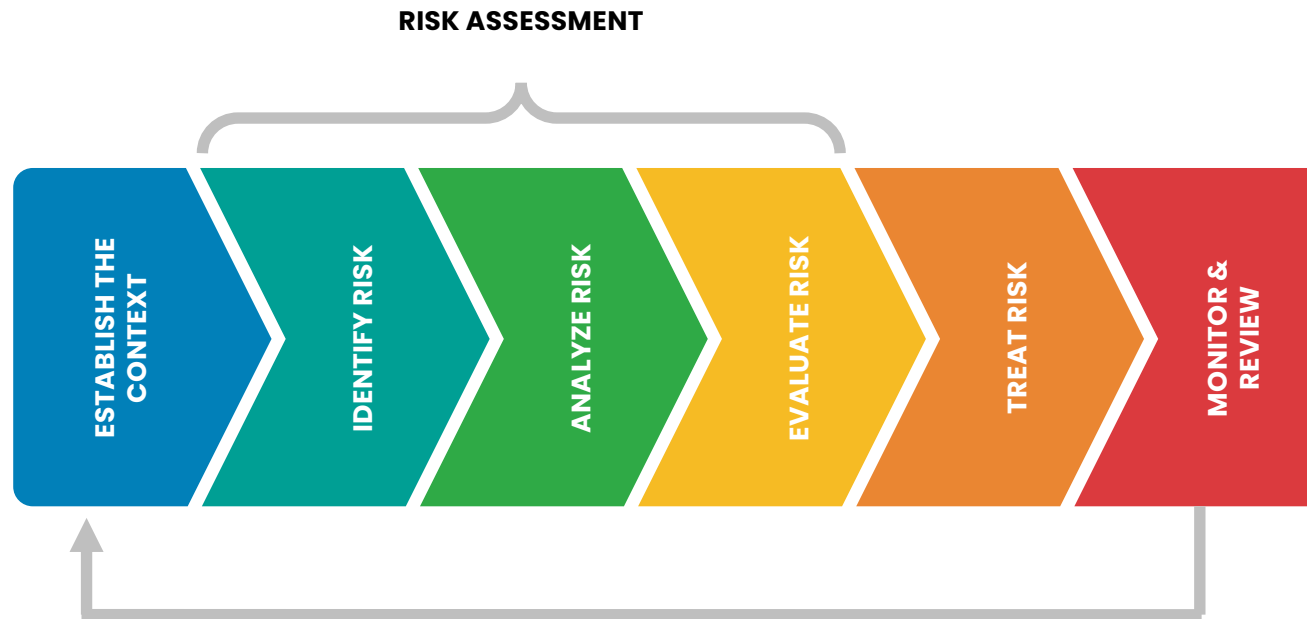
Risk Management Process

S.No.	RISK DESCRIPTION	RISK MITIGATION PLAN
1	Description.	Action
2	Description.	Action
3	Description.	Action





Risk Management Process



Information managed through **Risk Register**



Risk Management Process

MONITOR & REVIEW





Streamlined Work Processes, Improve Productivity

Work Processes Improvement



Work processes improvement and productivity

- Review the current workflow.
- Propose the areas to improve efficiency and productivity of the overall process.
- **Many methodologies** are applied to improve work-processes improvement.
- **Our focus is on AI Tools** usage to improve the workflow.



Case Studies

AI in Project Management

(Not in AI Project)

AI in Project Management



Case Study A

**AI in Project
Management**

Case Study: Detailed risk analysis table format with suggested solutions

Scenario: GenAI is used to generate the documentation.

Tools used: LLM via ChatGPT

Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Output was not in the existing format available in the company.
- Project Manager still needed time to review the outcome.

Solutions:

- Inhouse LLM (ChatGPT similar) model will help to resolve confidentiality issues.
- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Case Study B

**AI in Project
Management**

Case Study: Generation quality plan for a project.

Scenario: GenAI is used to generate Quality Plan for a project.

Tools used: LLM via ChatGPT

Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Project Manager still needed time to review the outcome.

Solutions:

- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Case Study C

**AI in Project
Management**

Case Study: Generation quality NCR reporting with analysis.

Scenario: AI Agent used to generate NCR (Non-Conformance Report) on a live system.

Tools used: LLM via ChatGPT

Concerns:

- Confidential information data could not be used.
- Need human intervention to review the outputs.
- Project Manager still needed time to review the outcome.

Solutions:

- Implement human-in-the-loop review for AI alerts
- Provide training to build trust among workers



Student Exercise 4

- Perform a research to find and list the available AI tools which may be used in the project management processes.
- Risk analysis and generating results.
- Quality plan and quality inspection checkpoints.
- Procurement delays impact on project.
- Any industry
- Any project

Prepare your list and submit via
Submission Link Provided Separately.



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End of class



Let the journey begin!

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